

DI – TABLES

Directions for questions 1 to 5: Answer these questions, on the basis of the information given below:

Grain \ Year	2015		2016		2017		2018	
	A	B	A	B	A	B	A	B
Rice	500	-100	400	-50	450	+50	600	+100
Wheat	600	+50	500	-20	450	-20	700	+50
Dal	300	-50	300	+10	250	-10	300	+20
Maize	200	-10	150	-10	150	0	200	+50

A: Estimated production (in million tonnes).

B: Comparison of the estimated production of a year with the previous year's actual production.

In the year 2014, what was the total production (in million tonnes) of all the four grains together?

☒ a) 1710 (Your answer - ✓)

☐ b) 2500

☐ c) 1490

☐ d) 1920

Solution:

The production of food grains in 2014 is as follows:

Rice → $500 + 100 = 600$

Wheat → $600 - 50 = 550$

Dal → $300 + 50 = 350$

Maize → $200 + 10 = \underline{210}$
 $\quad\quad\quad = \underline{1710}$

Choice (A)

If the estimated value for a particular grain for a single year is taken as an instance, then in the three years 2015, 2016 and 2017, in how many instances was the estimation correct?

☐ a) 4

☐ b) 1

☐ c) 3

☐ d) 5

Solution:

The estimation for Rice in 2016 and Maize in 2016 and 2017 were correct.
There are three such instances.

Choice (C)

What is the total production (in million tonnes) of Dal from 2015 to 2018?

- ☐ a) 2250
- ☐ b) 2190
- ☐ c) 1600
- ☐ d) Cannot be determined

Solution:

The production of Dal in 2018 can't be determined.
∴ The answer can't be determined for the given question.

Choice (D)

The production of maize in 2017 was what percentage of the production of rice in 2015?

☒ a) $33\frac{1}{3}\%$ (Your answer - ✓)

- ☐ b) 25%
- ☐ c) 26%
- ☐ d) $16\frac{2}{3}\%$

Solution:

Production of Maize in 2017 = 150

Production of Rice in 2015 = 450

The required percentage = $\frac{150}{450} \times 100 = 33\frac{1}{3}\%$

Choice (A)

During the period 2015 to 2017, in how many instances was the actual production greater than or equal to the estimated production (use data from previous questions if necessary)?

- ☐ a) 6 (Correct Answer - ✓)
- ☐ b) 3
- ☐ c) 4
- ☐ d) 5

Solution:

As per the given table, the actual production of the given food grains can be tabulated from 2014 to 2017 as follows.

	2014	2015	2016	2017
Rice	600	450	400	500
Wheat	550	520	470	650
Dal	350	290	260	280
Maize	210	160	150	150

Now by comparing this table and the given details, we can find that the estimated productions $\leq$ the actual production in the following instances:

Rice $\rightarrow$ 2016, 2017

Wheat $\rightarrow$ 2017

Dal $\rightarrow$ 2018

Maize $\rightarrow$ 2016, 2017

In six instances, the production of any one of the varieties was more than or equal to the estimated production.

Choice (A)

Directions for questions 6 to 10: Answer these questions, on the basis of the information given below:

The following table shows the score given by four judges J_1 to J_4 to five contestants C_1 to C_5 in the finals of a beauty contest.

Contestants Judges	C ₁			C ₂			C ₃			C ₄			C ₅		
	R ₁	R ₂	R ₃	R ₁	R ₂	R ₃	R ₁	R ₂	R ₃	R ₁	R ₂	R ₃	R ₁	R ₂	R ₃
J ₁	4	5	9	5	8	12	4	5	12	3	7	10	5	8	6
J ₂	6	7	12	7	10	7	8	6	10	8	4	7	8	8	12
J ₃	3	9	15	8	6	13	9	7	8	9	3	12	10	9	15
J ₄	8	8	14	9	5	18	6	9	15	6	5	15	3	7	13

The maximum score that can be awarded by a judge in round 1 (R_1) and round 2 (R_2) is 10 each, whereas the maximum score in round 3 (R_3) is 20. The contestant who scores the highest in any round is the winner in that round. The contestant who scores the highest in all the three rounds combined is the winner of the beauty contest.

Who is the winner of the beauty contest?

☐ a) C_1

☒ b) C_2 (Correct Answer - ✓)

☐ c) C_3

☐ d) C_5

Solution:

	C_1	C_2	C_3	C_4	C_5
R_1	21	29	27	26	26
R_2	29	29	27	19	32
R_3	50	50	45	44	46
Total	100	108	99	89	104

The winner of the contest is C_2

Choice (B)

Who is the runner-up in round 1 (R_1) of the beauty contest?

☐ a) C_2

☐ b) C_4

☐ c) C_5

☒ d) None of these (Correct Answer - ✓)

Solution:

	C_1	C_2	C_3	C_4	C_5
R_1	21	29	27	26	26
R_2	29	29	27	19	32
R_3	50	50	45	44	46
Total	100	108	99	89	104

The runner-up in round (R_1) of the beauty contest is C_3 .

Choice (D)

What is the highest percentage score secured by any of the five contestants in any of the given rounds?

- ☐ a) 62.5%
- ☐ b) 72.5%
- ☐ c) 75%

☐ d) 80% (Correct Answer - ✓)

Solution:

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	21	29	27	26	26
R ₂	29	29	27	19	32
R ₃	50	50	45	44	46
Total	100	108	99	89	104

The maximum possible score in R₁, R₂ and R₃ are 40, 40 and 80 respectively (∴ points given by the four judges are considered)

The highest score in R₁ is 29 won by C₂, which forms $\frac{29}{40} \times 100 = 72.5\%$ of the maximum possible.

Similarly, in R₂, the highest percentage = $\frac{32}{40}$
 $\times 100 = 80\%$.

Similarly, in R₃ the highest percentage

$$= \frac{50}{80} \times 100 = 62.5\%$$

Choice (D)

The average score awarded by J₃ in R₂ is approximately what percentage more/less than that awarded by J₂ in R₁?

☐ a) 8% less (Correct Answer - ✓)

- ☐ b) 9% more
- ☐ c) 10% less
- ☐ d) 11% more

Solution:

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	21	29	27	26	26
R ₂	29	29	27	19	32
R ₃	50	50	45	44	46
Total	100	108	99	89	104

The average score given by J₃ in R₂ = $\frac{9+6+7+3+9}{5} = 6.8$.

The average points allotted by J₂ in R₁ = $\frac{6+7+8+8+8}{5} = 7.4$

∴ The required percentage = $\frac{7.4 - 6.8}{7.4} \times 100 \approx 8\%$

Choice (A)

How many of the given contestants were given a higher score in R₃ than in the other two rounds combined, by at least two of the judges?

☐ a) 0 (Correct Answer - ✓)

☐ b) 2

☐ c) 3

☐ d) 5

Solution:

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	21	29	27	26	26
R ₂	29	29	27	19	32
R ₃	50	50	45	44	46
Total	100	108	99	89	104

None of the contestants were given a higher score in R₃ than in the other two rounds combined by at least two of the judges.

Choice (A)

Directions for questions 11 to 15: Answer these questions, on the basis of the information given below:

City	State	No. of. Schools	No. of arts colleges	No. of engineering colleges
Aizawl	Mizoram	410	46	12
Shillong	Meghalaya	500	44	14
Panaji	Goa	295	36	13
Hyderabad	Andhra Pradesh	493	24	12
Ranchi	Jharkhand	705	55	14
Dispur	Assam	780	28	11
Raipur	Chhattisgarh	872	59	12
Dehradun	Uttarakhand	315	48	14
Kolkata	West Bengal	608	53	13
Lucknow	Uttar Pradesh	870	31	12
Agartala	Tripura	725	39	13
Chennai	Tamil Nadu	772	45	11
Shimla	Himachal Pradesh	375	38	12
Bengaluru	Karnataka	324	25	12
Mumbai	Maharashtra	509	39	14
Patna	Bihar	637	52	10
Jaipur	Rajasthan	859	51	13
Chandigarh	Punjab	479	30	12

Directions for questions 11 to 13: Type in your answer in the input box provided below the question.

In how many cities is the number of engineering colleges greater than or equal to 12 and the number of schools less than or equal to 560?

Solution:

Observing the table, there are nine cities satisfying the given condition.
They are Aizawl, Shillong, Panaji, Hyderabad, Dehradun, Shimla, Bengaluru, Mumbai and Chandigarh.
Ans : (9)

Directions for questions 11 to 13: Type in your answer in the input box provided below the question.

Find the number of cities in which the number of arts colleges is between 20 and 35 (both inclusive) and the number of engineering colleges is 14.

Solution:

By observation, there are no cities satisfying the given condition.
Ans : (0)

Directions for questions 11 to 13: Type in your answer in the input box provided below the question.

Find the number of cities in which the number of arts colleges is in the range of 30 to 40 (both inclusive), the number of schools is in the range of 260 to 860 (both inclusive) and the number of engineering colleges not exceeding 12.

Solution:

The cities satisfying the requirements are Shimla and Chandigarh.

Ans : (2)

Directions for questions 14 and 15: Select the correct alternative from the given choices.

The city located in which state has the number of arts colleges greater than or equal to 35, the number of schools less than 360 and the number of engineering colleges less than 14?

- ☐ a) Uttarakhand
- ☐ b) Karnataka
- ☐ c) Himachal Pradesh

☐ d) Goa (Correct Answer - ✓)

Solution:

By observation, only Panaji in Goa has the number of arts colleges greater than 35 (36), the no. of schools less than 360 (295) and the no. of engineering colleges less than 14 (13).
Choice (D)

Directions for questions 14 and 15: Select the correct alternative from the given choices.

For which city is the ratio of the number of schools to the number of arts colleges, the highest?

- ☐ a) Lucknow
- ☐ b) Raipur
- ☐ c) Hyderabad
- ☐ d) Dispur

Solution:

By observation, Lucknow has the highest ratio.

Choice (A)